ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3, Q4

Question 1

(a) f_x = 3200 px. FOV = 160 mm. 1 pixel = 0.125 mm.

(b) Answer: 0.122 N.

(c) ZOH discretisation: $F = e^{AT}$, $G = \int_0^T e^{A\tau}B\, d\tau = A^{-1}(F - I)B$ $(A \neq 0)$.

Here $A = -2$, $B = 1$, $T = 0.1$: $F = e^{-0.2} = \mathbf{0.8187}$. $G = (F - 1)/A = (0.8187 - 1)/(-2) = \mathbf{0.0906}$.

$x_{k+1} = 0.8187\, x_k + 0.0906\, u_k$.

Question 3

(a) (1) Diferentiation is a high-pass operation (gain $\propto \omega$), so encoder measurement/quantisaton noise is amplified, by about $1/T = 1000$ at 1 kHz.

(2) Encoder quantisation: $\Delta\theta$ is a whole number of counts, so $\hat{\omega} = \Delta\theta/T$ moves in steps of (count size)/T; at low speed the estimat is a coarse staircase of zeros and spikes.

(b) $H(s) = \omega_c/(s + \omega_c)$, $\omega_c = 2\pi(20) = 125.66$ rad/s. Tustin: $s = (2/T)(z - 1)/(z + 1)$, T = 1 ms. With $a = \omega_c T/2 = 0.06283$: $H(z) = a(1 + z^{-1}) / [(1 + a) - (1 - a)z^{-1}]$.

$y_k = [a/(1+a)](u_k + u_{k-1}) + [(1-a)/(1+a)]\, y_{k-1} = \mathbf{0.05912\ (u_k + u_{k-1}) + 0.88177\ y_{k-1}}$.

DC gain: $z = 1$ gives $H(1) = 2(0.05912)/(1 - 0.88177) = 0.11824/0.11823 = 1.000$ ✓.

(c) $L = [18, 64]^T$.

(d) Advantage of the obsever: it uses the plant model and the applied input u, so its velocity estimate has (almost) no phase lag; the 20 Hz filter on differenced data always delays by about $1/(2\pi f_c) \approx 8$ ms. It can also reconstruct states that cannot be obtained by differencing (e.g. disturbance, when augmented).

Fliter is safer when the model is uncertain or changes (unknown payload, friction, contact, actuator saturation): the observer then gives a biased or poorly-behaved estimate while the model-free filter still follows the measured data.

Question 4

(a) Poles: $-2 \pm 2j$. $N_{ff} = 8$.

(b) $\theta_{ss} = 0.8$, error = 20%.

(c) Char. poly: $s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$.

Desiered: $(s + 4)(s^2 + 4s + 8) = s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 16s + 32$ (combining terms).

$2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$; $6 + k_1 = 16 \Rightarrow \mathbf{k_1 = 10}$; $K_I = \mathbf{32}$.

(d) Stedy state: $\dot{\xi} = 0$, so $r = \theta_{ss}$; the parameters b and c do not enter, hence zero error for any b and c as long as it is stable.

Failure mode: windup when the acuator saturtes. Remidy: reduce K_I to zero (or use a smaller gain) while saturated.